

Project Bibliography

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Contents

1	Pucci (2019) — Hedging the Treasury Lock	1
2	Pucci (2014) — Constant Maturity Treasury Convexity Correction	2
3	Pucci (2012a) — Constant Maturity Asset Swap Convexity Correction	3
4	Pucci (2012b) — Pricing Index-Linked Hybrids under a Local Volatility Market Model	4
5	Burgess — Bond Total Return Swaps: Theory, Pricing & Practice	4
6	Lou (2018) — Pricing Total Return Swap	5
7	Zhou (2008) — Bond Implied CDS Spread and CDS-Bond Basis	6
8	Axelsson & Renström (2013) — Credit-Linked Notes: Swedish Market Evidence	7
9	Anonymous — Treasury Lock Model	8
10	Anonymous — Why We Don't Discount at the "Risk-Free Rate"	8
11	Brigo & Morini (2005) — CDS Market Formulas and Models	9
12	Ametrano & Bianchetti (2013) — Multiple Interest Rate Curve Bootstrapping	10
13	Personal Work — The Treasury Lock: a comprehensive study	11

1 Pucci (2019) — Hedging the Treasury Lock

Identification

Title	Hedging the Treasury Lock
Author	Mario Pucci
Affiliation	Banca IMI SpA, Largo Mattioli 3, Milan, 20121, Italy
Email	mario.pucci@bancaimi.com
Date	19 August 2019
Source	ssrn:3386521
Local file	papers/pucci_2019_tlock/ssrn3386521.pdf
Length	25 pages

Abstract

The Treasury lock is a common pre-hedging derivative strategy the Street offers to their corporate clients. The paper provides a justification of the common practice of booking a short position in the Treasury lock as a forward contract on the underlying benchmark and a short position in the Then-Current Treasury lock as a forward contract on underlying benchmark rolled over the life of the contract.

Keywords

Treasury, lock, T-Lock, hedging, arbitrage-free, convexity, gamma, repo.

Contents

1. Introduction (Trade Description; Pricing the Standard T-Lock; Approximating a T-Lock as a Forward Contract).
2. Hedging the Standard T-Lock (Overhedge Error; Delta; Gamma; Replication via Repo).
3. Trading the Then-Current-T-Lock (Roll P&L; Forming the New Benchmark).
4. Statistical Analysis (Overhedge Error Estimation; Negative Gamma Bounds; Assessment of Roll Risk).
5. Valuation Adjustments.
6. Conclusions.
7. Appendices A–D: Simple Math; Key Derivatives; More Math; The Forward Price and the Repo.

References

22 numbered references. Notable: Hunt & Kennedy (*Financial Derivatives in Theory and Practice*, 2004); Hull (*Options, Futures and Other Derivatives*, 9th ed., 2015); Taleb (*Dynamic Hedging*, 1996); Apostol (*Calculus*, 1967); Carr & Madan (*Towards a Theory of Volatility Trading*, 2002); Andersen, Duffie & Song (*Funding value adjustments*, J. Finance 74:1, 2019); Adams & Smith (*Pre-Issuance Hedging of Fixed-Rate Debt*, J. Applied Corporate Finance 23:4, 2011); Pucci (*CMT Convexity Correction*, IJTAF 17:8, 2014); Ramirez (*Accounting for Derivatives*, Wiley 2015); Morini & Prampolini (*Risky Funding*, Landmarks in XVA 2016).

2 Pucci (2014) — Constant Maturity Treasury Convexity Correction

Identification

Title	Constant Maturity Treasury Convexity Correction
Author	Mario Pucci
Affiliation	Banca IMI SpA, Largo Mattioli 3, Milan, 20121, Italy
Email	mario.pucci@bancaimi.com
Dates	Received 13 Sep 2013; Revised 31 Mar 2014, 4 Jul 2014; Accepted 20 Aug 2014
Published	<i>International Journal of Theoretical and Applied Finance</i> 17(8), 2014
Source	ssrn:3387961
Local file	papers/pucci_2014_cmt/ssrn3387961.pdf
Length	15 pages

Abstract

In a Constant Maturity Treasury swap the exotic leg pays, for a given tenor, the yield-to-maturity computed out of a reference bond curve. This paper introduces a theoretical framework for the modelling of constant maturity treasury that takes into account default risk of bond issuer. As an application, we obtain, under simple but standard assumptions, analytical convexity corrections for some fundamental payoffs contingent on the constant maturity treasury.

Keywords

Convexity adjustment; default risk; constant maturity.

Contents

Sections (from document body):

1. Introduction.
2. Yield To Maturity and Constant Maturity Treasury.
3. Vanilla payoffs and the forward CMT.
4. Pricing framework (partial filtration).
5. Convexity correction under standard assumptions.
6. A hint on incorporating the smile.

References

Approximately 12 numbered references. Notable: Hagan (*Convexity Conundrums*, Wilmott 2003); Pelsser (*Mathematical Foundation of Convexity Correction*, Quantitative Finance 2001); Hunt & Kennedy (Wiley 2004); Boenkost & Schmidt (notes on convexity and quanto adjustments, 2003); Davies & Pugachevsky (Risk 2005).

3 Pucci (2012a) — Constant Maturity Asset Swap Convexity Correction

Identification

Title	Constant Maturity Asset Swap Convexity Correction
Author	Mario Pucci
Affiliation	Banca IMI, Milan
Email	mariopucci@hotmail.it
Date	27 February 2012 (revision timestamp)
Source	ssrn:1961545
Local file	<code>papers/pucci_2012_cmasw/ssrn1961545.pdf</code>
Length	10 pages

Abstract

As a consequence of the high volatility regime recently established in the government bond market, investors may seek hedging their exposure to floating asset swap spreads. We obtain an analytical convexity correction for the asset swap spread which is instrumental to the pricing of constant maturity asset swaps.

Keywords

Not stated explicitly. Topics: asset swap, convexity correction, CMS, constant maturity asset swap.

Contents

1. Introduction.
2. Constant Maturity Asset Swaps.
3. Definitions.
4. Pricing Framework.
5. One-Factor Linear Model.
6. Hedging Considerations.
7. Numerical Example.
8. Model Limitations.
9. Conclusions.

References

13 numbered references. Notable: Hagan (*Convexity Conundrums*, Wilmott 2003); Pelsser (*Mathematical Foundation of Convexity Correction*, Quantitative Finance 2001); Hunt & Kennedy (Wiley 2004); Zheng & Kwok (*Convexity meets replication*, J. Futures Markets 31, 2010); Nelson & Siegel (*Parsimonious modeling of yield curves*, J. Business 60, 1987); Lee & Wang (*Displaced Lognormal Volatility Skews*, Annals of Finance 2009); Choudhry (*CDS basis*, YieldCurve.com 2006); BIS (*Zero-coupon yield curves: technical documentation*, BIS Papers No 25, 2005).

4 Pucci (2012b) — Pricing Index-Linked Hybrids under a Local Volatility Market Model

Identification

Title	Pricing Index-Linked Hybrids under a Local Volatility Market Model
Author	Mario Pucci
Affiliation	Banca IMI, Milan
Email	mariopucci@hotmail.it
Date	22 February 2012 (with May 2012 revision)
Source	ssrn:2056277
Local file	papers/pucci_2012_hybrid/ssrn2056277.pdf
Length	5 pages

Abstract

We present a methodology to price a cash-settled swaption with strike contingent on the level of an equity index. The hybrid local volatility framework allows for calibration to market skew/smile and straightforward specification of an exogenous correlation. The framework can be extended to other multi-asset hybrid payoffs of European type.

Keywords

Not stated explicitly. Topics: hybrid product, local volatility, cash-settled swaption, equity-rates hybrid, index-linked, correlation.

Contents

Single-paper format, no explicit TOC. Discusses: hybrid product pricing problems (unobservable factors, correlation, smile); proposed local-volatility methodology; calibration to market skew/smile; extension to other European multi-asset hybrid payoffs.

References

6 numbered references including: Grzelak & Oosterlee (*An Equity-Interest Rate Hybrid Model With Stochastic Volatility and the Interest Rate Smile*, 2010); Piterbarg (*Smiling hybrids*, Risk, May 2006).

5 Burgess — Bond Total Return Swaps: Theory, Pricing & Practice

Identification

Title	Bond Total Return Swaps: Theory, Pricing & Practice
Author	Nicholas Burgess
Affiliation	Not stated in document
Date	Not stated on cover
Source	https://ssrn.com/author=1728976 ; companion repo at https://github.com/nburgessx/SwapsBook
Local file	papers/burgess_trs/5024091.pdf
Length	23 pages

Abstract

No abstract printed.

Keywords

Not stated. Topics: bond total return swap (TRS); contract specifications; constant units vs. constant notional; pricing formulae; worked case study.

Contents

Part One: Theory — Bond Total Return Swaps.

- Introduction to Bond TRS.
- Why Trade a TRS?
- Contract Specifications.
- Constant Units vs Constant Notional.

Part Two: Pricing & Practice — Case Studies.

- Pricing Formulae.
- Bond TRS Case Study.

References

No formal numbered reference list is printed. The document links to the author's SSRN author page (ssrn.com/author=1728976) and a companion GitHub repository (github.com/nburgessx/SwapsBook), and to the author's book *Low Latency Interest Rate Markets*.

6 Lou (2018) — Pricing Total Return Swap

Identification

Title	Pricing Total Return Swap
Author	Wujiang Lou
Affiliation	New York University, Courant Institute of Mathematics
Date	16 July 2018
Source	ssrn:3217420
Local file	<code>papers/lou_2018_trs/SSRNid32174202.pdf</code>
Length	25 pages

Abstract

Total return swap (TRS) involves a pricing dilemma: Libor discounting of its premium leg forces upfront payment of future funding premium, and yet replacing Libor with a firm's own funding rate falls into the well-known FVA debate trap. We consider TRS hedge financing from a repo market perspective and apply post-crisis derivatives valuation with collateralization and funding to TRS. We find that the financing cost of the TRS hedge should be reflected on the security leg, and the funding premium can only be discounted in conjunction with the TRS as a whole, depending on margining schemes. An easy to implement, recursive tree model is developed to value TRS with repo-style margining or defaultable underlying, together with any value adjustments.

Keywords

Total return swap, TRS pricing dilemma, funding costs, funding valuation adjustment, liability-side pricing, bond forward.

Contents

Topics (from inline section headings):

1. Introduction.
2. TRS mechanics (security leg, funding leg, repo financing).
3. Pricing under collateralisation and funding.
4. Recursive tree model for repo-style margining and defaultable underlying.
5. Value adjustments (XVA components).
6. Appendix: bond forward derivation under deterministic interest and repo rates.

References

References to the post-2008 funding / XVA literature including Andersen, Duffie & Song (FVA); Burgard & Kjaer; Piterbarg (*Funding beyond discounting*). Detailed list verified from the document body.

7 Zhou (2008) — Bond Implied CDS Spread and CDS-Bond Basis

Identification

Title	Bond Implied CDS Spread and CDS-Bond Basis
Author	Richard Zhou
Email	rzhou50@gmail.com
Affiliation	Not stated; disclaimer notes views are author's own
Date	15 August 2008
Source	ssrn:1265548
Local file	<code>papers/zhou_2008_cds/ssrn1265548.pdf</code>
Length	12 pages

Abstract

We derive a simple formula for calculating the CDS spread implied by the bond market price. Using no-arbitrage argument, the formula expresses the bond implied CDS spread as the sum of bond price, bond coupon and Libor zero curve weighted by risky annuities. We show that the bond implied CDS spread is consistent with the standard CDS pricing model if the survival probabilities and recovery are consistent with the bond price.

Keywords

Not stated explicitly. Topics: CDS spread, asset swap spread, CDS-bond basis, survival probability, recovery rate, bond-implied CDS.

Contents

1. Introduction.
2. CDS and Asset Swap Spreads (theory; main formulas; model applications).
3. Numerical examples.
4. Conclusions.
5. Appendix A: model derivation.
6. Appendix B: details.

References

Numbered list referencing standard credit-derivatives literature including: Davies & Pugachevsky on bond-implied CDS approximations; Berd et al. on survival-based modelling; the standard CDS pricing model literature; references on factors driving CDS-bond basis.

8 Axelsson & Renström (2013) — Credit-Linked Notes: Swedish Market Evidence

Identification

Title	On the pricing of Credit-Linked Notes: Evidence from the Swedish market
Authors	Marc Axelsson, Tobias Renström
Emails	40272@student.hhs.se, 40278@student.hhs.se
Affiliation	Stockholm School of Economics (SSE) — master's thesis; tutor: Michael Halling, Department of Finance
Date	15 May 2013
Source	No SSRN listed on document
Local file	<code>papers/axelsson_renstrom_2013_cln/cln_swedish_mkt.pdf</code>
Length	50 pages

Abstract

Based on international evidence of overpricing, we study the newly established credit-linked note (CLN) market in Sweden during the period 2012–2013. We derive a theoretical pricing model based on a reduced form framework and value the outstanding CLNs in the Swedish market. Our results suggest significant and consistent overpricing, ranging from 0.6% to 6.4%. Furthermore, when accounting for the counterparty credit risk exposure in the issuer, the overpricing increases substantially to a maximum of 8.5%. Given this considerable overpricing, it is possible that the counterparty risk is not priced in the CLNs. Plausibly, investors perceive the issuer as “too big to fail”, which is supported by previous findings of Arora et al. (2012). Lastly, our study indicates the existence of market frictions that influence the market pricing mechanism, obfuscating the relationship between observed prices and credit risk. Based on findings of Blanco et al. (2005), this could be a result of limited liquidity.

Keywords

Structured products, interest certificates, credit-linked notes, reduced form models.

Contents

Master's-thesis format (introduction, literature review, theory, data, empirical analysis, conclusion). Major sections expected: introduction; literature review (Burth et al. 2001; Stoimenov & Wilkens 2005; Blanco et al. 2005; Arora et al. 2012); reduced-form pricing model; data and methodology; empirical results (overpricing 0.6%–6.4% raw, up to 8.5% with CVA); discussion of market frictions and liquidity; conclusion.

References

Extensive thesis-style reference list. Notable: Burth, Kraus & Wuhrmann (Swiss structured product pricing, 2001); Stoimenov & Wilkens (German equity-linked structured products, 2005); Blanco, Brennan & Marsh (CDS-bond basis, 2005); Arora, Gandhi & Longstaff (counterparty risk and CDS spreads, 2012); Yiu (2012); Swedberg (2010).

9 Anonymous — Treasury Lock Model

Identification

Title	Treasury Lock Model
Author	Not stated (anonymous model documentation)
Affiliation	Document references https://finpricing.com/lib/FiZeroBond.html , suggesting FinPricing
Date	Not stated
Source	No SSRN/DOI on document
Local file	<code>papers/anon_treasury_lock_model/31670362145195.pdf</code>
Length	4 pages

Abstract

No abstract printed. The opening paragraph reads:

Treasury Lock (T-Lock) is a customized agreement that fixes the yield or price on a specified treasury security for a specific period. The buyer of the T-Lock is protected from a rise in the yield (i.e. a fall in price) of the underlying treasury security during the lock period. The period or term of the T-Lock is normally from one week to twelve months.

Keywords

Not stated. Topics: Treasury Lock, bond forward pricing, repo curve, clean/dirty price, locked yield, PV01.

Contents

1. Bond forward pricing formula (clean price form, repo discount factor).
2. Model inputs.
3. T-Lock NPV for clean-price lock.
4. T-Lock NPV for yield lock (with finite-difference PV01).
5. T-Lock NPV for dirty-price lock.
6. Note on PV01/NDELTA as approximate sensitivities.

References

One external link cited inline: <https://finpricing.com/lib/FiZeroBond.html>. No formal reference list.

10 Anonymous — Why We Don't Discount at the “Risk-Free Rate”

Identification

Title	Why We Don't Discount at the “Risk-Free Rate”
Subtitle	What Textbooks Get Wrong About Derivatives Pricing, and What Trading Desks Actually Do
Author	Not stated (anonymous)
Date	Not stated
Source	No SSRN/DOI on document
Local file	<code>papers/anon_discounting_textbooks/What_Textbooks_Get_Wrong_About_Discounting.pdf</code>
Length	9 pages

Abstract

No abstract printed. The opening paragraphs read:

If you studied derivatives pricing at university, you were almost certainly taught that derivatives are priced by taking the expected payoff under the risk-neutral measure and discounting at the risk-free rate. ... It's also, in its naïve form, wrong. ... The question "what rate do I discount at?" turns out to be far more subtle than the textbooks suggest, and the answer has reshaped the entire derivatives industry since 2008.

Keywords

Not stated. Topics: collateral rate, repo rate, funding rate, CSA, OIS, multi-curve, XVA (CVA/DVA/FVA), ColVA, partial CSA, cross-currency basis.

Contents

1. The General Framework.
2. The Textbook Argument, and Where It Hides Its Assumptions.
3. What the Rate Actually Represents: The Repo Rate (incl. equity, bond examples; summary table repo vs. collateral).
4. The CSA Revolution.
5. Collateral Regimes and the Discount Rate (perfect CSA; no CSA; partial CSA).
6. Linking the Collateral Rate to the Discount Rate (cash flow mechanics; martingale argument; worked swap example; cross-currency CSA; non-cash collateral; decision tree).
7. From Single-Curve to Multi-Curve.
8. The Correct Mental Model.

References

No formal reference list. Hull and Shreve mentioned as textbook examples.

11 Brigo & Morini (2005) — CDS Market Formulas and Models

Identification

Title	CDS Market Formulas and Models
Authors	Damiano Brigo, Massimo Morini
Affiliation	Brigo: Credit Models, Banca IMI, Milan. Morini: Università di Milano Bicocca
Date	September 2005
Source	Working paper, no SSRN ID printed on document
Local file	<code>papers/brigo_morini_2005_cdsmm/cdsmktfor.pdf</code>
Length	28 pages

Abstract

In this work we analyze market payoffs of Credit Default Swaps (CDS) and we derive rigorous standard market formulas for pricing options on CDS. Formulas are based on modelling CDS spreads which are consistent with simple market payoffs, and we introduce a subfiltration structure allowing all measures to be equivalent to the risk neutral measure. Then we investigate market CDS spreads through change of measure and consider possible choices of rates for modelling a complete term structure of CDS spreads. We also consider approximations and apply them to pricing of specific market contracts. Results are derived in a probabilistic framework similar to that of Jamshidian (2004).

Keywords

CDS, CDS option, market model, swap market model, subfiltration, change of measure, defaultable PV01, Black formula, CMCDS, constant maturity CDS, conditional independence.

Contents

1. Introduction.
2. Credit Default Swaps and Options (par CDS spread; CDS option payoff).
3. Standard Market Formula for CDS (numeraire $C_{a,b}$, change of measure, dynamics of $R_{a,b}$, empirical application).
4. Variables for a general CDS Market Model (one-period CDS forward rates; CDS spreads as forward conditional probability ratios; dynamics under different measures).
5. Dynamics and Constant Maturity CDS (Libor + CDS Market Model under independence; CMCDS pricing with drift-freezing approximation; empirical application on Fiat; alternative model with approximated payoff).
6. Conclusion.
7. Appendix (propositions and proofs).

References

Brigo (2004, 2005), Brigo & Mercurio (2001), Jamshidian (1997, 2004), Jeanblanc & Rutkowski (2000), Hull & White (2003), Schönbucher (1999, 2000, 2004), Bielecki & Rutkowski (2001), Wu (2005), and the seminal Black (1976) and Brace-Gatarek-Musiela (1997).

12 Ametrano & Bianchetti (2013) — Multiple Interest Rate Curve Bootstrapping

Identification

Title	Everything You Always Wanted to Know About Multiple Interest Rate Curve Bootstrapping But Were Afraid To Ask
Authors	Ferdinando M. Ametrano, Marco Bianchetti
Affiliation	Ametrano: Financial Engineering, Banca IMI, Milan. Bianchetti: Market Risk Management, Banca Intesa Sanpaolo, Milan
Date	2 April 2013 (first version 4 February 2013)
Source	ssrn:2219548 . Also a chapter in <i>Interest Rate Modelling After the Financial Crisis</i> (Bianchetti & Morini eds., Risk Books, 2013)
Local file	<code>papers/ametrano_bianchetti_2013_multicurve/ssrn2219548.pdf</code>
Length	82 pages

Abstract

After the credit and liquidity crisis started in summer 2007 the market has recognized that multiple yield curves are required for estimation of both discount and FRA rates with different tenors (e.g. Overnight, Libor 3 months, etc.), consistently with the large basis spreads and the wide diffusion of bilateral collateral agreements and central counterparties for derivatives transactions observed on the market... The theoretical pricing framework is introduced and modern pricing formulas for plain vanilla interest rate derivatives, such as Deposits, Forward Rate Agreements (FRA), Futures, Swaps, OIS, and Basis Swaps, are derived from scratch. The concrete EUR market case is worked out. . . The implementation of the proposed algorithms is available open source within the QuantLib framework.

Keywords

crisis, crunch, liquidity risk, funding risk, credit risk, counterparty risk, collateral, CSA, fixed income, interest rates, negative rates, Libor, Euribor, Eonia, yield curve, FRA curve, discount curve, single-curve, multiple-curve, bootstrapping, no-arbitrage, pricing, hedging, interest rate derivatives, Deposit, FRA, Futures, Swap, IRS, OIS, Basis Swap, turn of year, spline, greeks, sensitivity, QuantLib.

Contents

1. Introduction.
2. Classical and Modern Pricing & Hedging of Interest Rate Derivatives (single-curve vs. multiple-curve approach).
3. Notation and Basic Theoretical Framework (contract description, funding and collateral, pricing under collateral, interest rates, yield curves).
4. Bootstrapping Interest Rate Yield Curves (classical vs. modern bootstrapping; instruments selection — Deposits, FRA, Futures, IRS, OIS, Basis Swaps; synthetic instruments; interpolation; negative rates; endogenous vs. exogenous; turn of year effect; multiple deltas & hedging; performance and yield curve checks).
5. Implementation and Examples of Bootstrapping (Eonia ON, Euribor 1M / 3M / 6M / 12M, basis curves; full EUR market 11-Dec-2012 case).
6. Conclusions.
7. Appendices: pricing under collateral, Feynman-Kac, derivations of pricing formulas for FRA, Futures, IRS, OIS, IRBS.

References

Includes Mercurio (multiple-curve work), Brigo & Mercurio (*Interest-Rate Models*), Andersen & Piterbarg (*Interest Rate Modeling*), Hagan & West (curve interpolation), White (OpenGamma multi-curve), ISDA standard CSA documentation, plus the open-source QuantLib library which implements the paper's algorithms.

13 Personal Work — The Treasury Lock: a comprehensive study

Identification

Title	The Treasury Lock — a comprehensive study
Type	Personal-work integrated study (not a third-party paper)
Sources	(1) Anonymous practitioner T-Lock note; (2) Pucci 2019 (ssrn:3386521)
Date assembled	2026
Local file	<code>papers/personalWork/tlock_comprehensive/tlock_comprehensive.tex</code>
Length	10 pages

Abstract (own)

A unified treatment of Treasury Lock pricing combining the operational pricing recipe of the anonymous practitioner note (three lock conventions reduced to the bond forward formula) and the theoretical justification of Pucci (2019) (payoff identity, forward-contract overhedge, negative-gamma bound, repo replication, Then-Current roll). Two parts (Part I practitioner, Part II theoretical) plus a unified validation block reconciling the simple-vs-continuous compounding conventions on Pucci's 24-Jan-2019 worked example.

Keywords

T-Lock, repo, forward, convexity, gamma, pre-hedge, overhedge, Then-Current.

Contents

1. Introduction and Trade Description.
2. Unified Notation.
3. **Part I — Practitioner Pricing Model:** bond forward building block; three lock conventions; practitioner risks and sensitivities.
4. **Part II — Hedging and Convexity Analysis (Pucci 2019):** arbitrage-free pricing formula; forward proxy and overhedge; greeks (delta, gamma, sign conditions); repo replication; Then-Current T-Lock and roll risk; valuation adjustments; conclusions.
5. Validation & numerical examples (canonical 24-Jan-2019 case).

References

- Anonymous (n.d.), *Treasury Lock Model*, practitioner note.
- Pucci, M. (2019), *Hedging the Treasury Lock*, Banca IMI SpA, SSRN 3386521.